

DC-MoME-CRS: Full Methodology with Detailed Citations

Tổng quan

DC-MoME-CRS (Dialogue-Conditioned Mixture of Modality Experts for Conversational Recommender Systems) là một framework multimodal CRS mới, kết hợp 2 contributions chính: **(C1)** Dialogue-Conditioned Modality Expert Routing — chọn modality expert phù hợp cho mỗi entity tại mỗi turn dựa trên dialogue context, và **(C2)** Turn-Aware Routing with Momentum Drift — track sự tiến hóa của modality preference qua các turns và phát hiện topic shift. Framework được xây dựng trên nền tảng prompt learning cho CRS theo paradigm của UniCRS và mở rộng hướng multimodal của MSCRS. [1][2][3][4]

1. Notations và Task Definition

1.1 Notations

Ký hiệu	Ý nghĩa
$\mathcal{S} = \{s_1, s_2, \dots, s_M\}$	Tập hội thoại
$\mathcal{I} = \{i_1, i_2, \dots, i_N\}$	Tập items
$\mathcal{E} = \{e_1, e_2, \dots, e_K\}$	Tập entities, $\mathcal{I} \subseteq \mathcal{E}$
$s = \{c_1, c_2, \dots, c_B\}$	Một cuộc hội thoại gồm B turns (utterances)
$c_b = \{w_1, w_2, \dots, w_{l_b}\}$	Utterance thứ b gồm l_b tokens
$\mathcal{G}_{kg} = (\mathcal{E}, \mathcal{R})$	Knowledge Graph với relation set $\mathcal{R}$
$\mathbf{e}_i^m$	Representation của entity i ở modality $m \in \{kg, t, v\}$

1.2 Task Definition

Tại mỗi turn b trong cuộc hội thoại s , CRS thực hiện 2 subtasks:

- **Recommendation:** Xác định tập items $i \in \mathcal{I}$ phù hợp với user preference dựa trên conversation context $\{c_1, \dots, c_b\}$ và external knowledge.
- **Response Generation:** Sinh câu trả lời tự nhiên c_{b+1} , có thể chứa item được gợi ý.

2. Multimodal Encoding Layer

Tầng encoding trích xuất representations từ 3 nguồn thông tin khác nhau cho mỗi entity. Các encoder này follow thiết kế đã được validate trong các CRS trước đó.

2.1 Knowledge Graph Encoding (theo UniCRS, MSCRS)

Sử dụng R-GCN (Relational Graph Convolutional Network) trên DBpedia KG để encode structural relations giữa các entities. R-GCN mở rộng GCN tiêu chuẩn bằng cách tích hợp relation-specific weight matrices. Đây là encoder chuẩn đã được dùng trong UniCRS, MSCRS, VRICR, và STARCRS.[5][6][3][7][8][9][10][11]

Layer propagation rule:

$$\mathbf{e}_i^{kg,(l+1)} = \sigma \left(\mathbf{W}_0^{(l)} \mathbf{e}_i^{kg,(l)} + \sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_i^r} \frac{1}{c_{i,r}} \mathbf{W}_r^{(l)} \mathbf{e}_j^{kg,(l)} \right)$$

[12]

trong đó $\mathcal{N}_i^r$ là tập neighbors của entity i qua relation r , $c_{i,r}$ là hệ số chuẩn hóa, $\mathbf{W}_r^{(l)}$ là relation-specific weight matrix tại layer l . Để giảm tham số, dùng basis decomposition:[13][14]

$$\mathbf{W}_r^{(l)} = \sum_{b=1}^{B_{basis}} a_{rb}^{(l)} \mathbf{V}_b^{(l)}$$

[15]

với $\mathbf{V}_b^{(l)}$ là shared basis matrices và $a_{rb}^{(l)}$ là relation-specific coefficients.[6][8]

Output sau L layers: $\mathbf{E}_{kg} \in \mathbb{R}^{K \times d}$.

2.2 Textual Encoding (theo MSCRS)

Encode text description của mỗi item bằng pre-trained RoBERTa, tái sử dụng dữ liệu đã xử lý từ MSCRS:[2][7]

$$\mathbf{e}_i^t = \text{AvgPool}(\text{RoBERTa}(\text{desc}(i)))$$

[16]

Output: $\mathbf{E}_{txt} \in \mathbb{R}^{N \times d_t}$. Chỉ các entities là items mới có text description.[7]

2.3 Visual Encoding (theo MSCRS)

Encode image stills (poster, screenshots) bằng pre-trained ViT, tái sử dụng data từ MSCRS:[2][7]

$$\mathbf{e}_i^v = \frac{1}{l} \sum_{j=1}^l \text{ViT}(\text{img}_j^i)$$

[4]

Output: $\mathbf{E}_{vis} \in \mathbb{R}^{N \times d_v}$. Tương tự textual, chỉ items mới có visual features.[7]

2.4 Dialogue Turn-Level Encoding

Khác biệt so với UniCRS/MSCRS: encode **từng turn riêng** thay vì pool toàn bộ conversation. Điều này cần thiết cho Turn-Aware Routing (C2).

Sử dụng pre-trained LLM (Flan-T5-base hoặc LLaMA-2-7B + LoRA) làm backbone, theo xu hướng nâng cấp backbone trong ChatCRS và CRAVE:[17][18]

$$\mathbf{d}_b = \text{AttPool}(\text{LLM}(c_1, c_2, \dots, c_b))$$

[19]

trong đó AttPool là attention-weighted pooling:

$$\mathbf{d}_b = \sum_{j=1}^{p_b} \alpha_j \mathbf{h}_j^{(b)}, \quad \alpha_j = \frac{\exp(\mathbf{w}^\top \mathbf{h}_j^{(b)})}{\sum_k \exp(\mathbf{w}^\top \mathbf{h}_k^{(b)})}$$

[20]

Sequence of turn representations: $\mathbf{D}_s = [\mathbf{d}_1, \mathbf{d}_2, \dots, \mathbf{d}_B] \in \mathbb{R}^{B \times d_c}$.

3. Modality Alignment (Preliminary Step — cite STARCERS, CLIP)

Trước khi routing, cần đảm bảo 3 modality representations nằm trong semantic space tương đồng, để gating network nhận input có thể so sánh được. Bước này follow approach đã được validate trong STARCERS và CLIP.[9][10][21]

3.1 Projection vào Common Space

Project mỗi modality về common space $\mathbb{R}^d$ bằng MLP projectors:

$$\mathbf{h}_i^{kg} = \text{MLP}_{kg}(\mathbf{e}_i^{kg}), \quad \mathbf{h}_i^t = \text{MLP}_t(\mathbf{e}_i^t), \quad \mathbf{h}_i^v = \text{MLP}_v(\mathbf{e}_i^v)$$

[22]

Mỗi MLP gồm: Linear $\rightarrow$ GELU $\rightarrow$ Linear $\rightarrow$ LayerNorm. Thiết kế này theo projection network của STARCERS.[9]

3.2 Cross-Modal Contrastive Alignment

Sử dụng InfoNCE loss để align representations của cùng entity across modalities. Theo STARCERS, dùng KG representation làm anchor:[21] [23][9]

$$\mathcal{L}_{kg-txt} = - \sum_{i \in \mathcal{I}_b} \log \frac{\exp(\mathbf{h}_i^{kg\top} \mathbf{h}_i^t / \tau)}{\sum_{j \in \mathcal{I}_b} \exp(\mathbf{h}_i^{kg\top} \mathbf{h}_j^t / \tau)}$$

[24]

$$\mathcal{L}_{kg-vis} = - \sum_{i \in \mathcal{I}_b} \log \frac{\exp(\mathbf{h}_i^{kg\top} \mathbf{h}_i^v / \tau)}{\sum_{j \in \mathcal{I}_b} \exp(\mathbf{h}_i^{kg\top} \mathbf{h}_j^v / \tau)}$$

[25]

$$\mathcal{L}_{align} = \mathcal{L}_{kg-txt} + \mathcal{L}_{kg-vis}$$

[2]

với τ là temperature scalar, $\mathcal{I}_b$ là batch of items.

Framing trong paper: *"Following STARCRS (Yuan et al., 2026) and CLIP (Radford et al., 2021), we pre-align modality representations to ensure the gating network receives semantically comparable inputs. This is a prerequisite for effective expert routing, not a standalone contribution."*[10][21]

4. DC-MoME: Dialogue-Conditioned Modality Expert Routing (C1)

Đây là **contribution chính thứ nhất**. Khác biệt cốt lõi so với tất cả existing work:

Method	Routing type	Context-aware?	Per-entity?
MSCRS[2]	Mean fusion	✗ Static	✗
STARCRS[10]	Sigmoid gating	✗ Static (gate only depends on entity features)[9]	✗
FuseMoE[26]	Top-k token routing	✗ Input-only[27]	✗ Per-token
I ² MoE[28]	Interaction-type routing	✗ Based on modality interaction type	✗
HM4SR[29]	Temporal MoE	Partial (timestamp-based)[30]	✗ Per-item in sequence
DC-MoME (Ours)	Dialogue-conditioned routing	✓ Dialogue semantics	✓ Per entity per turn

4.1 Modality Expert Networks

Mỗi modality có một dedicated expert network, lấy cảm hứng từ MoE expert design trong FuseMoE nhưng specialized cho per-entity level: [31][26]

$$\mathbf{o}_i^m = \text{Expert}_m(\mathbf{h}_i^m), \quad m \in \{kg, t, v\} \quad [32]$$

Mỗi expert gồm:

$$\text{Expert}_m(\mathbf{x}) = \text{LayerNorm}(\mathbf{W}_2^m \cdot \text{GELU}(\mathbf{W}_1^m \cdot \mathbf{x} + \mathbf{b}_1^m) + \mathbf{b}_2^m) \quad [33]$$

với $\mathbf{W}_1^m \in \mathbb{R}^{d_{ff} \times d}$, $\mathbf{W}_2^m \in \mathbb{R}^{d \times d_{ff}}$. Expert FFN có shared architecture nhưng **separate parameters** — mỗi expert học transform riêng cho modality của mình.

4.2 Dialogue-Conditioned Cross-Attention Gating

Sử dụng dialogue turn representation $\mathbf{d}_b$ (Eq. 5) làm **query** để attend lên entity modality features, xác định modality nào relevant nhất cho entity i tại turn b :

Bước 1 — Tính attention score giữa dialogue context và mỗi modality:

$$a_i^{m,(b)} = \frac{(\mathbf{W}_Q \mathbf{d}_b)^\top (\mathbf{W}_K^m \mathbf{h}_i^m)}{\sqrt{d_k}}, \quad m \in \{kg, t, v\}$$

[34]

trong đó $\mathbf{W}_Q \in \mathbb{R}^{d_k \times d_c}$ là query projection (shared across modalities), $\mathbf{W}_K^m \in \mathbb{R}^{d_k \times d}$ là modality-specific key projection.

Bước 2 — Softmax để có routing weights:

$$\mathbf{g}_i^{(b)} = \text{softmax}([a_i^{kg,(b)}, a_i^{t,(b)}, a_i^{v,(b)}]) \in \mathbb{R}^3$$

[35]

Ý nghĩa: dialogue context "hỏi" mỗi modality "*bạn có relevant không?*". Khi user nói "*I want something visually stunning*", visual key sẽ có alignment cao với dialogue query $\rightarrow g_i^v$ tăng. Khi user nói "*directed by Christopher Nolan*", KG key (chứa director relation) sẽ align tốt hơn $\rightarrow g_i^{kg}$ tăng.

So sánh kỹ thuật với STARCRCR: STARCRCR dùng static sigmoid gate $g = \sigma(W \cdot e_{kg} + b)$ — gate weights **chỉ phụ thuộc entity features**, không thay đổi theo dialogue context. DC-MoME gate thay đổi theo **mỗi turn** vì $\mathbf{d}_b$ thay đổi.[9]

So sánh với FuseMoE: FuseMoE dùng standard top-k gating $G(x) = \text{TopK}(\text{softmax}(W_g \cdot x))$ — chỉ dựa trên input token features, không có external conditioning signal. DC-MoME conditioning trên dialogue context — một tín hiệu bên ngoài entity.
[36][31]

5. Turn-Aware Routing with Momentum Drift (C2) — Core Novelty

Đây là **contribution chính thứ hai**, tách biệt hoàn toàn khỏi mọi existing work. Motivation: user modality preference **không nhảy ngẫu nhiên** giữa các turns mà có **momentum** (xu hướng tiếp tục) và **drift** (thay đổi khi topic shift).

So sánh chi tiết với related temporal MoE works:

Aspect	TIME-MoE (ICLR 2026)[37][38]	HM4SR (WWW 2025)[29][30]	DC-MoME + Drift (Ours)
Domain	Sensor/medical/sentiment	Sequential rec	Conversational rec
Temporal signal	Sensor readings + timestamps; RUS decomposition (Redundancy/Uniqueness/Synergy)[39]	Interaction timestamps $\rightarrow$ interval embedding + absolute time embedding[40]	Dialogue turn semantics
Routing basis	Temporal RUS sequences $\rightarrow$ RUS-aware router with attention + GRU[39]	Gating on temporal embedding concatenated with item features[40]	Dialogue content + routing history + topic shift
Expert type	Regular + Synergy experts (cross-attention)[39]	Interactive MoE + Temporal MoE (hierarchical)[40]	Modality experts (KG/Text/Visual)
Temporal granularity	Continuous time lags τ [37]	Discrete timestamps[30]	Discrete dialogue turns
Per-entity ?	No (per-token)[38]	No (per-item in sequence)[30]	Yes — per entity per turn

5.1 Routing History Tracking

Lưu routing weights từ tất cả turns trước cho mỗi entity i :

$$\mathcal{G}_i^{(<b)} = \{\mathbf{g}_i^{(1)}, \mathbf{g}_i^{(2)}, \dots, \mathbf{g}_i^{(b-1)}\}$$

[41]

5.2 Modality Preference Momentum via EMA

Tính momentum vector thông qua Exponential Moving Average — recent turns có trọng số lớn hơn:

$$\mathbf{g}_i^{mom,(b)} = \beta \cdot \mathbf{g}_i^{mom,(b-1)} + (1 - \beta) \cdot \mathbf{g}_i^{(b-1)}$$

[42]

với $\beta \in (0, 1)$ là decay factor (hyperparameter). Initialization:

$\mathbf{g}_i^{mom,(1)} = [1/3, 1/3, 1/3]$ (uniform). EMA là kỹ thuật chuẩn trong signal processing, nhưng **chưa ai áp dụng cho modality routing history**.

5.3 Topic Shift Detection

Tính mức độ semantic shift giữa turn hiện tại và turn trước thông qua cosine distance:

$$\Delta_b = 1 - \frac{\mathbf{d}_b^\top \mathbf{d}_{b-1}}{\|\mathbf{d}_b\| \|\mathbf{d}_{b-1}\|}$$

[43]

$\Delta_b \approx 0$: topic ổn định $\rightarrow$ momentum nên được giữ. Δ_b lớn: topic shift $\rightarrow$ nên giảm momentum influence và tin tưởng current turn routing hơn.

5.4 Learned Drift Gate

Thay vì hard threshold, sử dụng **learned gate** λ_b để tự động cân bằng giữa current routing và historical momentum:

$$\lambda_b = \sigma(\mathbf{w}_\lambda^\top [\mathbf{d}_b; \Delta_b \cdot \mathbf{1}] + b_\lambda)$$

[17]

trong đó σ là sigmoid, $\mathbf{w}_\lambda \in \mathbb{R}^{d_c+1}$, $\Delta_b \cdot \mathbf{1}$ broadcast scalar thành vector. λ_b gần 1 $\rightarrow$ tin current turn (xảy ra khi topic shift lớn). λ_b gần 0 $\rightarrow$ tin momentum (xảy ra khi topic ổn định).

5.5 Drift-Aware Final Routing

Kết hợp current routing (Eq. 14) với momentum (Eq. 16), điều chỉnh bởi drift gate (Eq. 18):

$$\hat{\mathbf{g}}_i^{(b)} = \lambda_b \cdot \mathbf{g}_i^{(b)} + (1 - \lambda_b) \cdot \mathbf{g}_i^{mom, (b)} \quad [44]$$

5.6 Fused Entity Representation

$$\mathbf{e}_i^{fused, (b)} = \hat{g}_i^{kg, (b)} \cdot \mathbf{o}_i^{kg} + \hat{g}_i^{t, (b)} \cdot \mathbf{o}_i^t + \hat{g}_i^{v, (b)} \cdot \mathbf{o}_i^v \quad [18]$$

Entity representation **thay đổi mỗi turn** — cùng entity "The Dark Knight", ở turn 1 representation có thể thiên về KG (actors, director), ở turn 3 thiên về visual (IMAX scenes).

5.7 Missing Modality Handling

Entities không phải item (actors, directors, genres) không có text description hay images. Re-normalize routing weights:

$$\hat{g}_i^{m, (b)} \leftarrow \frac{\hat{g}_i^{m, (b)} \cdot \mathbb{1}[m \text{ available for } i]}{\sum_{m'} \hat{g}_i^{m', (b)} \cdot \mathbb{1}[m' \text{ available for } i]} \quad [45]$$

6. Load Balancing Loss

Theo Shazeer et al. (2017), thêm load balancing loss để tránh expert collapse — hiện tượng 1 expert luôn nhận weight cao nhất trong khi các expert khác bị bỏ qua:[46][47]

$$\mathcal{L}_{bal} = 3 \sum_{m=1}^3 f_m \cdot p_m \quad [48]$$

trong đó:

- $f_m = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \mathbb{1}[\arg \max(\hat{\mathbf{g}}_i^{(b)}) = m]$ — fraction of entities trong batch routed to expert m
- $p_m = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \hat{g}_i^{m,(b)}$ — average routing probability for expert m

Multiplier 3 (= số experts) theo chuẩn hóa trong Switch Transformers.[49][36]

7. Unified Adaptive Prompt Framework (Application — cite UniCRS)

Follow prompt learning paradigm của UniCRS, nhưng thay thế static entity representations bằng dynamic fused representations từ DC-MoME.[3][5][1]

7.1 Entity Selection

Chọn entities xuất hiện trong conversation s đến turn b :

$$\mathbf{V}_s^{(b)} = \{\mathbf{e}_j^{fused,(b)} \mid j \in \text{entities}(s, \leq b)\} \in \mathbb{R}^{q \times d}$$

[50]

7.2 Cross-Attention Entity-Dialogue Fusion

Theo UniCRS, dùng cross-attention để inject entity information vào dialogue representation:[3]

Cho Recommendation: Entity features attend to dialogue tokens:

$$\tilde{\mathbf{V}}_s = \text{CrossAttn}(\mathbf{V}_s^{(b)}, \mathbf{T}_s) \in \mathbb{R}^{q \times d}$$

[51]

Cho Conversation: Dialogue tokens attend to entity features:

$$\tilde{\mathbf{T}}_s = \text{CrossAttn}(\mathbf{T}_s, \mathbf{V}_s^{(b)}) \in \mathbb{R}^{p \times d_c}$$

[52]

7.3 Recommendation Prompt và Prediction

Construct soft prompt theo UniCRS format:[3]

$$\mathbf{P}_{rec} = [\tilde{\mathbf{V}}_s; \mathbf{O}_{rec}; \text{tokens}(s)]$$

[53]

trong đó $\mathbf{O}_{rec} \in \mathbb{R}^{n_{soft} \times d_c}$ là learnable task-specific soft tokens.

Prediction:

$$\hat{\mathbf{r}}_s = \text{MeanPool}(\text{LLM}(\mathbf{P}_{rec}; \theta_{frozen}))$$

[54]

$$P(i \mid s) = \text{softmax}(\hat{\mathbf{r}}_s^\top \mathbf{E}_{fused}^{(b)}[\mathcal{I}])$$

[55]

7.4 Conversation Prompt và Generation

$$\mathbf{P}_{conv} = [\tilde{\mathbf{T}}_s; \mathbf{O}_{conv}; \text{tokens}(s)]$$

[56]

Response generation qua autoregressive decoding:

$$P(w_t \mid w_{<t}, s) = \text{LLM}(\mathbf{P}_{conv}; \theta_{frozen})$$

[57]

8. Training Pipeline

8.1 Training Phases

Phase	Mục tiêu	Loss	Trainable components	Freeze
Phase 0: Alignm ent	Align modality spaces	$\mathcal{L}_{align}$ (Eq. 10)	MLP projectors	LLM, RGCN, RoBERTa, ViT
Phase 1: Entity Pre- trainin g	Link prompt to entity context (theo UniCRS[3])	$\mathcal{L}_{pre} + \delta\mathcal{L}_{bal}$	Expert FFNs, router, drift module, soft tokens	LLM backbone
Phase 2: Recom mendat ion	Predict ground-truth items	$\mathcal{L}_{rec} + \delta\mathcal{L}_{bal}$	Expert FFNs, router, drift module, soft tokens	LLM backbone
Phase 3: Conver sation	Generate responses	$\mathcal{L}_{gen} + \delta\mathcal{L}_{bal}$	Expert FFNs, router, drift module, soft tokens	LLM backbone

8.2 Loss Functions

Phase 1 — Entity Pre-training (theo UniCRS):[3]

$$\mathcal{L}_{pre} = - \sum_{s \in \mathcal{C}_b} \sum_{e \in \mathcal{E}_s} \log P(e | s) + \delta\mathcal{L}_{bal}$$

[58]

Phase 2 — Recommendation:

$$\mathcal{L}_{rec} = - \sum_{s \in \mathcal{C}_b} \log P(i^* | s) + \delta\mathcal{L}_{bal}$$

[59]

trong đó i^* là ground-truth recommended item.

Phase 3 — Conversation (teacher forcing):

$$\mathcal{L}_{gen} = - \sum_{s \in \mathcal{C}_b} \sum_{t=1}^{|c_{b+1}|} \log P(w_t \mid w_{<t}, \mathbf{P}_{conv}) + \delta \mathcal{L}_{bal}$$

[60]

8.3 Backbone Options

Option	Model	Total param s	Trainabl e params	GPU requirem ent
A (recomm ended)	Flan-T5-base (250M)	~265M	~15M	1×A6000 48GB
B (strong)	Flan-T5-large (780M)	~800M	~20M	1×A100 40GB
C (stretch)	LLaMA-2-7B + LoRA (r=16)	~7.1B	~30M	1×A100 80GB

LLM backbone frozen — chỉ train expert FFNs, router, drift module, soft tokens, projectors. Approach này theo parameter-efficient paradigm đã validate trong UniCRS và DCRS.[61][3]

8.4 Hyperparameters

Parameter	Ý nghĩa	Gợi ý	Nguồn
τ	Temperature cho contrastive	0.07	CLIP[21], STARCRS[9]
β	EMA decay cho momentum	0.5	Standard EMA
δ	Weight cho load balancing	0.01	Shazeer et al. [46]
d_{ff}	Expert FFN hidden dim	512	—
d_k	Cross-attention key dim	64	—
n_{soft}	Số soft tokens	20	UniCRS[3]
Learning rate	—	1e-4 (AdamW)	MSCRS[7]
Batch size	—	256 (rec), 64 (gen)	MSCRS[7]

9. Bản đồ Provenance: Mỗi component từ đâu

Component	Equation(s)	Source	Role in paper
R-GCN encoding	Eq. 1-2	Schlichtkrull et al. 2017[8]; UniCRS[3]; MSCRS[7]	Existing — cite
RoBERTa text encoding	Eq. 3	MSCRS[7][2]	Existing — cite
ViT visual encoding	Eq. 4	MSCRS[7][2]	Existing — cite
AttPool dialogue encoding	Eq. 5-6	Standard technique	Adapted for turn-level
MLP projection + InfoNCE	Eq. 7-10	CLIP[21]; STARCRS[9]	Preliminary — cite STARCRS
Expert FFN per modality	Eq. 11-12	FuseMoE[31][26] concept	Adapted for modality level
Cross-attention gating	Eq. 13-14	□ NOVEL	Core C1
Routing history + EMA	Eq. 15-16	□ NOVEL	Core C2
Topic shift detection	Eq. 17	□ NOVEL	Core C2
Learned drift gate	Eq. 18-19	□ NOVEL	Core C2
Weighted expert fusion	Eq. 20	Standard MoE output[46][36]	Standard — cite

Component	Equation(s)	Source	Role in paper
Missing modality re-norm	Eq. 21	Standard technique	—
Load balancing loss	Eq. 22	Shazeer et al. 2017[46]; Switch Transformer[49]	Standard — cite
Soft prompt framework	Eq. 23-30	UniCRS[3][5]	Existing — cite
Training phases	Eq. 31-33	UniCRS[3]; MSCRS[7]	Adapted

10. Positioning Statement cho Paper

"Recent works have explored multimodal fusion for CRS through static semantic graphs (MSCRS, Wei et al. 2025) and contrastive alignment with adaptive gating (STARCRS, Yuan et al. 2026). In parallel, temporal MoE architectures have been proposed for sequential recommendation (HM4SR, Zhang et al. 2025) and sensor fusion (TIME-MoE, Han et al. 2025). However, conversational recommendation presents a fundamentally different temporal signal: the evolving semantic content of dialogue turns. We are the first to (1) condition modality expert routing on dialogue semantics per entity per turn, and (2) model the momentum and drift of modality preferences across conversational turns, enabling entity representations that dynamically adapt to what the user cares about at each moment in the conversation."[37][29][10][2]

References

1. [Towards Unified Conversational Recommender Systems via Knowledge-Enhanced Prompt Learning](#)
2. [MSCRS: Multi-modal Semantic Graph Prompt Learning Framework ...](#) - In this paper, we propose a multi-modal semantic

- graph prompt learning framework for CRS, named MSCR...
3. [\[2206.09363\] Towards Unified Conversational Recommender ...](#) - Conversational recommender systems (CRS) aim to proactively elicit user preference and recommend hig...
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 6. [Relational Graph Convolutional Networks \(RGCN\) - Emergent Mind](#) - Relational Graph Convolutional Networks (RGCN) extend GCNs to multi-relational, directed graphs usin...
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 12. [report.docx](#) - # CRS

Luồng CRS cơ bản

Conversation $\rightarrow$ entity $e \rightarrow$ (external knowledge) $\rightarrow$ entity representation E .

co...

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30. [Hierarchical Time-Aware Mixture of Experts for Multi-Modal...](#) - The proposed HM4SR method effectively addresses the challenges in multi-modal sequential recommendat...
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- 본 논문은 대화형 추천 시스템(Conversational Recommender Systems, CRS)을 위한 새로운 프레임워크인 MSCRS(Multi-modal Semantic ...
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- 50. [Leveraging Large Language Models in Conversational Recommender Systems](#)
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- 55. [Collaborative Retrieval for Large Language Model-based ...](#) - CRAG is the first approach that combines state-of-the-art LLMs with CF for conversational recommenda...
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60. [Why Multi-Interest Fairness Matters: Hypergraph ...](#) - ACL Anthology - We proposed a novel framework, Hypergraph Contrastive Multi-Interest Learning for Fair Conversationa...
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